

Mengti Sun

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EDUCATION

M.S.E. Robotics

UNIVERSITY OF PENNSYLVANIA

Philadelphia, PA | May 2023

B.A. Computer Science, B.A. Cognitive Science

UNIVERSITY OF CALIFORNIA, BERKELEY

Berkeley, CA | Dec 2020

WORK EXPERIENCE

AMAZON | SOFTWARE DEVELOPMENT ENGINEER II

Seattle, WA | Aug 2026 – PRESENT

Amazon Robotics

- Design and develop **adaptable multimodal data collection pipelines** for robotics research, enabling synchronized capture of manipulation, vision, and sensor data for training robotic systems
- Develop **sensorized manipulator interfaces and gripper systems** that capture human grasp demonstrations across diverse objects, integrating force, motion, and end-effector sensing for robot learning

AMAZON | SOFTWARE DEVELOPMENT ENGINEER I

Seattle, WA | Jan 2024 – Aug 2026

Fulfillment By Amazon (FBA)

- Designed and developed services to identify defects in the **Fulfillment by Amazon (FBA)** inbound process, implementing coaching, fee charging, or dispute options for global third-party sellers to mitigate future issues
- Integrated 6-sided item images captured by robotic scanners at stow stations, implementing and optimizing CNN and Mask R-CNN algorithms on **AWS SageMaker**, increasing defect identification accuracy by **20%**

AMAZON | SOFTWARE DEVELOPMENT ENGINEER INTERN

Seattle, WA | May 2022 – Aug 2022

- Implemented automation for reviewing ASIN scheduling information for merchandising products in Amazon Style showrooms and reduced curation time from **214 days** to **a few hours** for a single store
- Developed backend APIs with **Kotlin** and frontend UI with **React** for assortment management; implemented **AWS CDK** stacks for **multi-region deployment** and **cross-region calls**

BYTEDANCE | SOFTWARE DEVELOPMENT ENGINEER INTERN

Beijing, China | Apr 2021 – Aug 2021

- Optimized the smart education service QPS by largely modifying the logic of message queue, cache, and database
- Pioneered microservices for order management, quality inspection and spot checks at **API and RPC** levels, enhancing efficiency by **50%** in manual recommendation and spot checks

RESEARCH EXPERIENCE

VISION-AND-PHYSICS-BASED OBJECT RECONSTRUCTION

PHILADELPHIA, PA | JAN 2024

Dynamic Autonomy and Intelligent Robotics (DAIR) Lab at University of Pennsylvania

- Implemented and evaluated an instance-agnostic **RGB-D 6D pose tracking** pipeline for unfamiliar rigid objects without requiring category- or instance-level **3D shape priors**, including experiments involving fast object motion and substantial occlusion
- Integrated vision-based pose estimates with the ContactNets **contact-dynamics learning** framework, eliminating reliance on **AprilTag-based pose tracking** and enabling joint estimation of object geometry, pose trajectories, and physical behavior from RGB-D video and robot proprioception

OPTIMIZED JUMPING THROUGH HOOPS WITH QUADRUPEDAL ROBOTS

BERKELEY, CA | JUN 2021

Hybrid Robotics Lab at UC Berkeley

- Formulated a **trajectory optimization** algorithm that determines gait sequences, step timings, footholds, swing-leg motions and 6D body motions for **MIT mini cheetah robots** to jump through hoops and avoid obstacles (**CASE '21**)
- Generated in **real time** (within a few seconds) the complete sequences of motions based on a desired goal state and the number of steps and successfully demonstrated feasibility on a real robot

SKILLS

Languages: C/C++, Python, Java, Golang, Matlab, TypeScript

Technology: PyTorch, Tensorflow, AWS, OpenCV, ROS, CUDA, Spring Boot, GraphQL, MySQL, MongoDB, DevOps